

Optimal Strategy — 2026 *FIRST* Global Challenge: *Igniting Innovation*

A strategic analysis of the 2026 FGC game manual, derived from the scoring rules in §3 and confirmed against the bundled `score-calculator.html`.

1. Scoring at a glance

For each Regional Alliance:

```
score = ceil( suppression_balls × climb_multiplier
              + partner_climbs × 25
              + extinguisher_balls
              + coopertition_bonus
              + foul_bonus_from_opponent )
```

where

```
climb_multiplier = 1.00 + Σ per-robot increment
                  Contact = 0.05
                  Zone 1  = 0.10
                  Zone 2  = 0.20
                  Zone 3  = 0.30
```

Action	Pts	Notes
Wildfire in Suppression Unit	$1 \times \text{multiplier}$	Regional only
Wildfire in Extinguisher	1	Both alliances
Partner Climb	25 / supported robot	Regional, max 2
Coopertition Bonus	10 / 25 / 40	Global, needs 4 / 5 / 6 robots in Zone 3
Foul against opponent	+5% / +10% of their pre-penalty score	Awarded, not deducted

Ball pool: **500 wildfire** released onto the field at “go”. Match length: **2:30**.

2. The single most important number

The climb multiplier compounds on every suppression ball. With three robots in Zone 3, the multiplier reaches **1.90** — every suppression ball is worth **90 % more** than its face value.

Climb result (3 robots)	Multiplier	Value of a suppression ball
All None	1.00	1.00
All Contact	1.15	1.15
All Zone 1	1.30	1.30
All Zone 2	1.60	1.60
All Zone 3	1.90	1.90

Combined with two partner climbs (50 pts), a successful triple Zone 3 climb is worth roughly $+0.90 \times \text{suppression_balls} + 50$ vs. doing nothing. On 80 suppression balls that’s **+122 pts** — typically larger than the entire ball-scoring contribution of the other alliance.

Conclusion: the climb is not a finishing flourish, it is the centerpiece of the match. Build, drive, and budget time around it.

3. Strategic priorities (in order)

Priority 1 — Triple-Zone-3 climb every match

- Designate **one robot as the climber/anchor**: tall, rigid, with hooks or hangers that can carry two other robots' weight.
- The other two robots must have **mating geometry** to be hooked/supported by the anchor: lift bars, ring loops, or shaped tops.
- Robots must enter the brace in Zone 1 (rule G15) and traverse along it, so the anchor needs a passive winch or pull-up mechanism reaching 197 cm.
- Because the brace deflects up to 100 mm at center, anchors should over-engineer for sag.

Priority 2 — Lock in the *Coopertition* Bonus

- With both alliances at 3-in-Z3 → 6 robots → **+40 pts to every team**.
- This is the single best return-on-coordination in the game. **Talk to the opposing alliance before each match**. It is a “global alliance” element by rule — there is no cost to helping them succeed at climbing.
- Even a partial agreement (one alliance climbs 3, the other 1 or 2) earns +10 or +25 universally. **Never refuse to participate**.

Priority 3 — Suppression > Extinguisher (when you climb)

Compare per-ball value to your alliance:

Destination	Without climb	With Zone-3 triple
Suppression Unit	1.00	1.90
Extinguisher	1.00	1.00

Every match in which you intend to climb, **the Suppression Unit dominates**. The Extinguisher only “wins” per ball if your alliance has no climb, and even then it ties.

But: Extinguisher points go to **both** alliances, so they’re “free” from a *regional rivalry* perspective — they don’t help you beat the other side. They do help your raw match score, and the ranking is the **average of your best matches**, not a head-to-head record.

Heuristic: target roughly **80 % of robot ball-handling time on Suppression, 20 % on routing balls into Fire-Shield ports for the human to feed the Extinguisher**. Drop Extinguisher entirely if you’re behind in tempo and need to maximize multiplier-amplified Suppression.

Priority 4 — Direct suppression over fire-shield routing

A robot may score Suppression two ways: 1. Pick up wildfire → carry to Suppression Unit (single transport). 2. Push wildfire into Fire-Shield port → human places in chute → human pulls lever → ball drops to floor → robot picks up → carry to Suppression Unit (double transport + serial dependence on human).

Path 1 is strictly faster per ball. **Path 2 only makes sense** when: - A robot can shove but not lift; the human + chute acts as a “ball lift”. - The chute is geometrically lined up with a robot that can intake from above.

Most teams should design their robot to handle path 1 natively.

Priority 5 — Ranking-aware play

- Ranking score = **average of best matches** (lowest one dropped — rule M21, §6.3).
- Tiebreaker 1: **highest single match score** → there is a real incentive to go for a big match rather than steady mediocre matches.
- Tiebreaker 2: **cumulative Suppression points** → another reason to bias toward Suppression over Extinguisher.

Implication: don't sandbag. A blowout score with a good climb is more valuable than two safe medium scores.

4. The 2:30 timeline

A workable match clock for an alliance executing the full plan:

Time (remaining)	Phase	What each robot does
2:30 → 2:00	Rush	All 3 robots grab loose wildfire from center, hammer Suppression. Human players push port-shoved balls to the Extinguisher in parallel.
2:00 → 1:00	Sustained scoring	Continue Suppression. Robots cycle near Suppression Unit; humans empty Fire Shields whenever a port fills.
1:00 → 0:35	Position	Anchor robot drives toward Zone 1 of the brace. Other two finish last cycles and converge on the brace.
0:35 → 0:10	Climb	Anchor enters Zone 1 → Zone 2 → Zone 3. Partners attach to anchor and are lifted.
0:10 → 0:00	Hold	All three stable. Verify no contact with the playing-field surface; no contact with brace support hardware (rule G16/G19 notes — only the steel pipe counts).

Why so early on the climb? The brace climb is the highest-variance scoring action in the game. Rule G15: lose contact with Zone 2/3 → **must restart from Zone 1**. A failed climb at 0:15 leaves no time to recover. Starting the climb at 0:35 buys one retry.

5. Penalty-aware design (don't bleed points)

Foul penalties are calculated as **% of the offending alliance's pre-penalty score and added to the opponent** — so a foul on a high-scoring match hurts much more in absolute points than on a low-scoring one. A 10 % MAJOR FOUL on a 300-point match = **+30 pts to opponent**.

Highest-risk fouls to design / drive away from:

- **G01 / M16** — Pinning, blockading, tipping → MAJOR FOUL. Drive defensively around opponents on the Suppression Unit approach; don't camp the opposing robots' lanes.

- **G13** — Launching wildfire through the air → **RED CARD**. If you use any shooter mechanism, ensure trajectories stay low; carry-and-drop is safer.
- **G18** — Knocking another alliance’s robot off the brace → escalating cards. Climb lanes are shared with your own partners; teach drivers to wait their turn rather than push.
- **M14** — If your robot doesn’t move at the start, your 30-second touch window is the only repair window. Have a “dead-robot checklist”: power, plug, loose wire. Don’t touch any other robot or game piece while doing it (M12).

6. Robot design implications (translating strategy → hardware)

Strategy point	Design requirement
Triple Zone-3 climb	One anchor robot with a deployable hanging mechanism reaching ~200 cm; passive lock so power loss doesn’t drop it. Two partners with rigid, defined “hook here” geometry.
Direct Suppression scoring	Vertical scoring height: clear the 165 cm canopy bottom edge by a margin (R03 says ± 25 mm field tolerance).
High wildfire throughput	Wide intake (R05 lets you extend 50 cm in one horizontal direction post-start), high-capacity hopper (G12: no holding limit).
Reliable start	R04: $50 \times 50 \times 50$ cm starting volume. Bias compact and robust; extensions deploy <i>after</i> “go”.
AprilTag navigation	Five tags on Suppression Units + Extinguisher faces — auto-aim Suppression scoring saves driver bandwidth for late-game maneuvering.
Easy game-piece removal (R08)	Make sure a stuck wildfire can be removed without power; otherwise a jam = game over.

7. Alliance-level coordination

Three teams form a Regional Alliance every match in Ranking play, randomly assigned (§3.2). You won’t have practice time together. Before each match, agree on three things at the queue:

1. **Climber identity** — which robot is the anchor? Pick by mechanism, not by ego. If no robot can anchor a triple climb, plan a double climb and one robot scoring to the buzzer.
2. **Lane assignment** — left robot, center robot, right robot. Three robots converging on one Suppression Unit causes congestion.
3. **Extinguisher commitment** — agree on whether human players will work the Extinguisher at all this match. If yes, robots route a fraction of balls to Fire Shield ports rather than Suppression.

Cross-alliance: **always pitch the 6-robot Cooptition climb** to the opposing alliance. The +40 bonus is uncorrelated with who wins the match — it is pure additive score for everyone. Refusing it is leaving free ranking points on the table.

8. Pareto-optimal match outcomes (back-of-envelope)

Assume the field of 500 balls is split roughly evenly: ~200 Suppression each side, ~100 Extinguisher (the rest end up uncontained — G14: those score 0).

Floor — no climb, decent scoring: - $150 \text{ Suppression} \times 1.00 + 0 + 100 + 0 = \mathbf{250}$

Solid — triple Zone 1 + two partner climbs: - $180 \text{ Suppression} \times 1.30 + 50 + 100 + 0 = \mathbf{384}$

Strong — triple Zone 3 + two partner climbs (alone): - $200 \text{ Suppression} \times 1.90 + 50 + 100 + 10 = \mathbf{540}$ (*Coop bonus from 4 robots if opponent gets 1 in Z3*)

Maxed — both alliances triple-Z3 (full coopertition): - $200 \times 1.90 + 50 + 100 + 40 = \mathbf{570}$

Note the last two rows: cooperating with the opposing alliance to get the full +40 bonus *strictly Pareto-improves* your score ($570 > 540$) — there is no defensive reason to deny it.

9. TL;DR — the 7-point strategy

1. **Build to climb.** The Zone-3 multiplier is the single biggest scoring lever in the game.
2. **One climber, two hangers.** Design the alliance composition around partner climbs; +50 pts is essentially free if the geometry mates.
3. **Start the climb at 0:35.** Failed climbs need restart time (rule G15).
4. **Score Suppression directly**, not through Fire Shields. Bypass the chute.
5. **Help the Extinguisher only in parallel** — humans work it while robots stay on Suppression.
6. **Always pitch the 6-robot coopertition climb** to the opposing alliance. +40 pts to both sides is the highest ROI conversation in the queue.
7. **Don't foul.** Penalties scale with your match score, so they punish your best matches the hardest, which are the ones determining your ranking (lowest match is dropped per M21).

The team that wins is the team whose climb works the most reliably. Everything else is volume.